Geochemistry and paleomagnetism of Late Cretaceous mafic dikes in Kerala, southwest coast of India in relation to large igneous provinces and mantle plumes in the Indian Ocean region

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ABSTRACT

New geochemical and paleomagnetic results are presented on two Late Cretaceous dikes of the 85–90 Ma leucogabbroic and doleritic dikes and the 65–70 Ma dolerites in Kerala, India. The dikes are rich in incompatible elements, have fractionated patterns with light rare-earth element enrichment and are akin geochemically to Cretaceous basalts on the east coast of Madagascar. The magmas were formed at garnet lherzolite depths above the Marion plume, constituting part of a large igneous province in Madagascar. In contrast, the 65–70 Ma dolerites are moderately depleted in incompatible elements, with almost flat, rare-earth element patterns and resemble the upper formations of the Deccan Traps and the tholeiitic dikes of the Seychelles. These dolerites were formed by melting of spinel lherzolite over the Reunion plume. Paleomagnetic data from the dikes and the other coeval igneous units from south India provide the 90 Ma pole (latitude: 24°; longitude: 293°; A_{95} = 5.9; N = 18 sites) for India. The 65–70 Ma dolerites possess both normal and reverse polarities, and the mean pole (latitude: 36°; longitude: 283°; A_{95} = 5.7°; N = 10 sites) compares well with the Deccan superpole. Paleolatitude estimates indicate ~5° southward migration for the Marion plume and a northward migration for the Reunion plume, in conformity with global mantle-circulation models; however, distinguishing migration of the Reunion plume from the effects of true polar wander is difficult. Furthermore, the geodynamic reconstructions extending the shear zones of southern Madagascar into south India are not tenable.

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INTRODUCTION

The intraplate magmatism that constitutes large igneous provinces (LIPs) of Late Cretaceous age in the Indian Ocean region is attributed to mantle plume activity. The Deccan flood basalts in western India constitute a major LIP of 65–70 Ma age that is generally linked to Reunion hotspot magmatism (Morgan, 1981; Courtillot et al., 2003), although a nonplume origin is also proposed (Sheth, 2005). During the past decade, isotopic dating revealed occurrences of mafic dikes of Deccan age in coastal central and north Kerala (Fig. 1) in southwestern India (Radhakrishna et al., 1994; Radhakrishna et al., 1999) and also in the Goan region (Widdowson et al., 2000). In addition, another magmatic episode of 85–90 Ma of limited areal spread has been recognized in the coastal regions of southwestern India. It is represented by dike intrusions in central and north Kerala, Agali-Coimbatore sector (Radhakrishna et al., 1994, 1999; see Fig. 1) and two isolated dike occurrences in the interior Karnataka craton (Kumar et al., 2001) and St. Mary's Islands volcanism (Torsvik et al., 2000; Pande et al., 2001; Melluso et al., 2009). Yet, no major large igneous province of this age is known from India. However, precise Ar-Ar and U-Pb isotope dating (Storey et al., 1995; Torsvik et al., 1998; Melluso et al., 2005) has identified widespread flood basalt volcanism of this age in Madagascar. This ca. 90 Ma magmatism in Madagascar and India has been attributed to the plume-head stage of the Marion plume (Storey et al., 1995; Ernst and Buchan, 1997).

We present new major- and trace-element data, oxygen isotope results, and paleomagnetic data, for the mafic dikes of these two age groups exposed in Kerala. We then used the data to draw correlations of these dikes in relation to the age data, with certain basalts erupted in the Cretaceous Madagascar and Deccan igneous provinces. The results for the (85–90 Ma) dikes, combined with other data, are used to constrain the Marion plume magmatism in India. The paleomagnetic interpretations are further used to test the predicted migration of mantle plumes in Indian Ocean region by the global mantle-circulation models (O'Neill et al., 2003) and suggested, in conformity with the predictions, ~5° southward migrations for the Marion plume.

GEOLOGICAL SETTING, AGE, AND PETROGRAPHY

The central and north Kerala basic dikes intrude Archean high-grade, granulite-facies metamorphic rocks, mostly charnockites with subordinate local occurrences of garnet-bearing gneiss and basic granulite. In central Kerala, a 20–50-m-thick, NNW-SSE–trending, en echelon leucogabbroic dike is exposed over a distance of more than 100 km (Fig. 1), and is cut by 65–70 Ma, NW-SE–trending dolerite intrusions. Again, a similar leucogabbro dike is exposed in north Kerala extending for over 50 km, and the dolerites occur mainly in a set of mutually orthogonal directions of NW-SE and NE-SW. The dolerites are fine to medium grained, 5–10 m thick, and crop out in a broad zone >50 km wide and >100 km long. It is difficult to estimate the lengths of individual dikes because of the thick laterite soil cover. However, from discontinuous exposures, individual dikes may extend to a maximum length of ~10 km. On the southern flanks of Nilgiri massif, in the Agali-Coimbatore region, the western part of Bhavani shear zone, the prominent swarm is of Proterozoic dolerites (Radhakrishna et al., 1995), but subsidiary Phanerozoic dikes also occur as ESE-WSW thin (5–6 m), gray, vesicular dolerites within the biotite/hornblende-quartz-feldspar gneiss country rock.

GSA Bulletin; January/February 2012; v. 124; no. 1/2; p. 240–255; doi: 10.1130/B30288.1; 10 figures; 1 table; Data Repository item 2011165.

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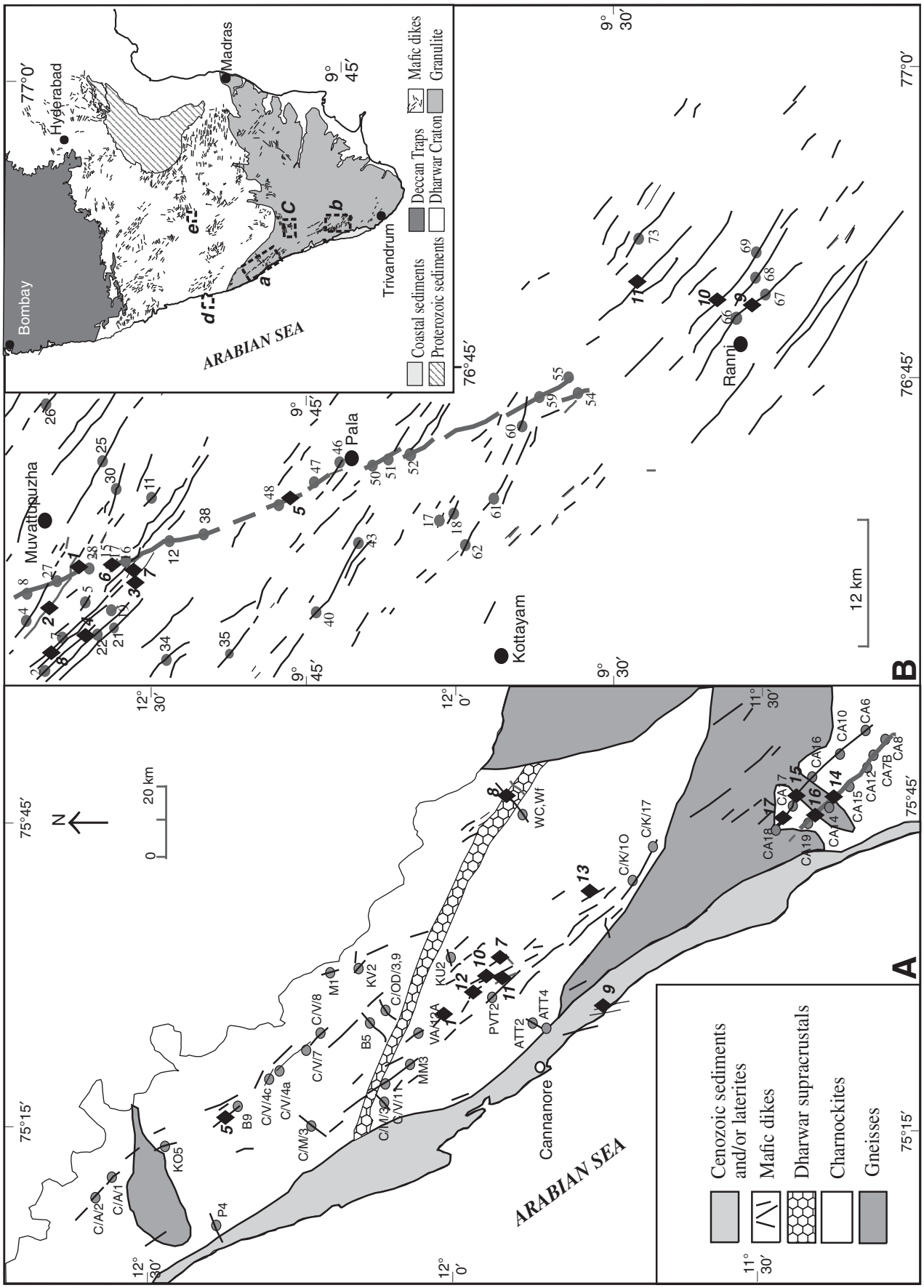


Figure 1. Geological map showing the distribution of Cretaceous basic dikes in (A) north Kerala and (B) central Kerala, India (compiled from Geological Survey of India maps). See the inset map for the other locations discussed in the text. In the inset map: (a) north Kerala, (b) central Kerala, (c) Agali-Coimbatore, (d) St. Mary's Islands volcanic rocks, and (e) Cretaceous dikes of Karnataka. Solids circles are geochemical samples, and diamonds are paleomagnetic sites.

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K-Ar analyses of four central Kerala and two north Kerala leucogabbro whole-rock samples yielded mean ages of 81.3 ± 0.7 Ma and 81.7 ± 1.4 Ma, respectively (1σ error; Radhakrishna et al., 1994; Radhakrishna et al., 1999). $^{40}Ar/^{39}Ar$ age spectra of the Cretaceous dike, with similar geochemistry, from Agali-Coimbatore area indicate 90 Ma as the maximum age. Thus, the K-Ar age may correspond to the cooling stage of the dike, and the actual crystallization age of the dike could be placed in the 85–90 Ma range (Radhakrishna et al., 1999). This age correlates well with the $^{40}Ar/^{39}Ar$ age for a dike in Dharwar craton (Kumar et al., 2001), $^{40}Ar/^{39}Ar$ and U-Pb ages obtained for St. Mary's Islands volcanic rocks (Torsvik et al., 2000; Pande et al., 2001) and the mean $^{40}Ar/^{39}Ar$ and U-Pb ages for Madagascan basalts and rhyolites (Storey et al., 1995; Torsvik et al., 1998; Melluso et al., 2005).

Three of the central Kerala dolerite samples yielded geologically meaningful isochron ages for temperature increments >680 °C (1σ errors, see Radhakrishna et al., 1994), while plagioclases from three dolerites of north Kerala yielded plateau ages (Radhakrishna et al., 1999); and while one age is 59.7 ± 5.3 Ma, the lowest and highest ages are 68.0 ± 2.2 Ma and 70.2 ± 1.4 Ma, respectively. The ages obtained are similar to the $^{40}Ar/^{39}Ar$ age of alkalic igneous rocks in the northern part of the Deccan province (e.g., the Mer Mundwara complex and Sarnu-Dandali intrusions, Rajasthan, both 68.5 ± 0.2 Ma; Basu et al., 1993), the mean $^{40}K/^{40}Ar$ age of 67.4 ± 1.0 Ma of the Jawahar Formation in the northern Deccan province (Chenet et al., 2007), and the K-Ar ages of two basic dikes from Praslin Island in the Seychelles (73.5 ± 2.2 Ma and 69.1 ± 1.7 Ma; Dickin et al., 1986). Thus the $^{40}Ar/^{39}Ar$ ages probably are close to the true crystallization ages of the dikes. Certain Deccan lavas appear to have erupted at ca. 68–69 Ma, although the great majority of these basalts give ages of 66–65 Ma (e.g., Baksi, 1987; Duncan and Pyle, 1988; Venkatesan et al., 1993; Hofmann et al., 2000), and relatively minor activity persisted ca. 60–61 Ma (Sheth et al., 2001). A recent compilation of published $^{40}Ar/^{39}Ar$ ages and the new $^{40}K/^{40}Ar$ ages (Table 2; Chenet et al., 2007) clearly identifies these three volcanic pulses in Deccan province. Reunion-like geochemistry is known from Bibai volcanics from Quetta-Zhob area, Pakistan (Mahoney et al., 2002); however, their ^{40}Ar-^{39}Ar isotope age (73.2 ± 1 Ma) is older than the ages of Deccan Trap basalts.

Petrography

The 85–90 Ma coarse-grained gabbroic dike samples are composed of plagioclase, clinopyroxene, and opaque minerals. The samples are subophitic with large plagioclase grains showing a wide range in composition. Plagioclase ranges from An_{63} to An_{27} in basaltic to basaltic andesite samples and from An_{59} to An_{19} in andesite samples (e.g., CK-1h). In Agali-Coimbatore dolerites, it shows a limited range ($An_{64–62}$). Zoning is rare, and variation is limited between cores and rims. Quartz and apatite are frequent accessory minerals. Apatite occurs as euhedral to subhedral and sometimes contains inclusions of small tear-shaped quartz and thin flakes of thread-like biotite. Micropegmatitic texture is common along the margins of large plagioclase crystals. Clinopyroxene is anhedral to subhedral, ranging from augite to ferroaugite in composition ($Mg_{56–21}Fe_{42–16}Ca_{42–32}$). Pigeonite ($Mg_{57–45}Fe_{47–34}Ca_{14–5}$) is a rare phase. Deuteric alteration to pale-green amphibole (actinolite) or to biotite is common along grain boundaries or cleavage planes. However, subhedral biotite and green amphibole form accessory phases in the more evolved samples of leucogabbroic dike in north Kerala. Many samples show patchy replacement by clay. Fe-Ti oxides occur as dispersed grains. Titanomagnetite (X_{usp} = 0.95–0.99) and ilmenite (X_{ilm} = 0.96–0.99) constitute the oxide phases. Titanite is a rare accessory phase in Agali-Coimbatore dikes.

The 65–70 Ma dolerites are fine to medium grained and contain plagioclase, clinopyroxene, and Fe-Ti oxides. Olivine and orthopyroxene are rare accessory phases and occur as euhedral crystals and granular aggregates. The rocks are quite fresh, showing ophitic to subophitic intergrowths of plagioclase and clinopyroxene within the ophitic matrix. Plagioclase phenocrysts and rare glomerocrysts are set in a matrix of interlocking plagioclase, pyroxene, and Fe-Ti oxides. Plagioclase is andesine-labradorite in composition ($An_{69–37}$), with more calcic cores. Rarely the cores may range to bytownite (up to An_{82}) in central Kerala dolerites. Clinopyroxene is common as microphenocrysts and also constitutes the groundmass. It predominantly is augite to ferroaugite in composition ($Mg_{52–34}Fe_{49–12}Ca_{39–17}$), but ranges occasionally to magnesian pigeonite ($Mg_{57–49}Fe_{50–35}Ca_{9–1}$) or pigeonite ($Mg_{47–32}Fe_{56–52}Ca_{16–1}$). Micropegmatitic intergrowths of quartzofeldspathic material occur occasionally along plagioclase grain boundaries. Opaque mineralogy includes Fe-Ti oxides and sulfides that are common as dispersed grains, often with skeletal outlines. Fe-Ti oxides generally range from titanomagnetite (X_{usp} = 0.86–0.99) to ilmenite (X_{ilm} = 0.92–0.99). Some grains show typical intergrowths of magnetite and ilmenite. Garnet ($Alm_{62}gro_{19}pyr_{9}$) was noticed in one north Kerala dolerite sample (OD3), and it may constitute a xenocryst phase.

SAMPLES AND ANALYTICAL METHODS

Samples for geochemical study were collected from the 85–90 Ma dikes of central and north Kerala and Agali-Coimbatore sector and the 65–70 Ma dikes of central and north Kerala (Fig. 1). Samples weighing 1–2 kg were collected avoiding top lateritized surfaces, and 97 samples (33 from 85–90 Ma dikes and 64 from 65–70 Ma dikes) without serious alterations are selected for chemical analysis. Generally the 65–70 Ma dolerite samples are very fresh (weight loss on ignition, LOI <1%). The 85–90 Ma dike samples may have undergone minor to moderate alteration (LOI: up to 2.5 wt%). Fourteen sites (85 samples) were sampled for paleomagnetic investigations. These include eleven sites (dikes) of 65–70 Ma dolerites in north Kerala and three sites (20 samples) of 85–90 Ma dikes (two from leucogabbroic dikes in north Kerala and one from a dolerite of the Agali-Coimbatore sector). At least six block samples were collected from each site using a magnetic compass for orientation. Paleomagnetic results from eleven 65–70 Ma dolerites and three sites of 85–90 Ma leucogabbro from central Kerala were published earlier (Radhakrishna et al., 1994). These data are combined with the new results to present a total account of paleomagnetism of the Cretaceous dikes in Kerala.

Major oxides were analyzed by the X-ray fluorescence methods at the National Geophysical Research Institute (NGRI), Hyderabad; Geological Survey of India, Bangalore; Center for Earth Science Studies, Trivandrum; and University of Leicester, UK. Trace elements were determined by X-ray fluorescence (XRF) for most of the samples at NGRI and University of Leicester. These include Cr, Ni, Rb, Sr, Ba, Zr, Zn, Y, and V; also Th, Co, Nb, Ga, La, Ce, and Nd were analyzed in 17 samples at the University of Leicester. Rare-earth elements, along with other trace elements (Rb, Sr, Ba, V, Zn, Th, Zr, Y, Nb, Ga, Sc, Cu, Hf, and Ta) were determined for 43 representative samples by inductively coupled plasma–mass spectrometry at NGRI following the methods of Balaram et al. (1990). Seven samples of the 85–90 Ma dikes and six 65–70 Ma dolerite samples were analyzed for oxygen isotope analysis at the Stable Isotope Laboratory, University of Georgia, USA. Repeated analyses of an internal quartz sand sample (RQS) gave errors of ± 0.30‰ (2σ) on a $\delta^{18}O$ value of +17.5‰. Two 85–90 Ma dolerite samples from Agali-Coimbatore were analyzed for Sr and Nd isotopes along with the Proterozoic dikes from the same area at the Department of Terrestrial Magnetism, Carnegie Institution of Washington, USA. The

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mass spectrometer measurements were as described previously (Radhakrishna et al., 1995). Samples were chosen to provide the best possible areal coverage of the dikes. Major- and trace-element data with estimates of precision and accuracies are given in GSA Data Repository Tables DR1 and DR2[1].

The natural remanent magnetization (NRM) was measured on a MOLSPIN spinner magnetometer. All samples were treated by alternate field (AF) demagnetizations until a single component remained (generally up to 40–50 mT and rarely beyond), and 30 samples were treated by thermal demagnetizations. The characteristic remanent magnetization (ChRM) directions were delineated using orthogonal projections and principal component analysis using computer programs (Paleomagnetism Analysis Programs [PMGSC], version 3 developed by Randy J. Enkin, Geological Survey of Canada; http://www.gsc.nrcan.gc.ca/sw/paleo_e.php) accepting linear trajectories of consecutive points with angular uncertainty <10°. Temperature-susceptibility measurements were carried out for thermomagnetic analysis using a Bartington temperature-susceptibility system interfaced with a computer.

RESULTS

Geochemistry

The 85–90 Ma dike samples are quartz-normative tholeiites. The dikes from central Kerala and Agali-Coimbatore range from basalt to basaltic andesite on the total alkali-silica diagram of Le Maitre (2002; figure not shown). Two samples (CK-59 and CK-16b) extend into the andesitic field. The dike samples in north Kerala mostly range from basaltic andesite to andesite, the latter being more predominant. Significant alteration and/or contamination is evident in a few samples (CK-50, -59, -52, and -28, and NK-19; see Table DR1 [see footnote 1]) that have strikingly high SiO_2 (53.2–61.4 wt%), K_2O (1.4–2.5 wt%), and Rb values (38–95 ppm), sawtooth-shaped, mantle-normalized patterns and prominent negative Nb anomalies ($Nb^* = Nb_n/\sqrt{(Th_n * La_n)} = 0.2–0.4$; subscript n denotes primordial-mantle–normalized value). Even excluding these altered and/or contaminated samples, the dikes are rich in Rb (10–50 ppm), Sr (400–870 ppm), Ba (170–600 ppm), and P_2O_5 (0.35–0.70 wt%). Their Ni and Cr contents are low (10–50 ppm and 8–119 ppm, respectively), and Y and Zr abundances vary from 25 to 79 ppm and 200 to 560 ppm, respectively (Table DR1 [see footnote 1]). Zr/Y values range from 4.6 to 10.7, whereas Zr/Nb = 11–25 and La/Nb = 1.0–4.3. Figure 2A shows that the 85–90 Ma dikes are enriched in the light rare-earth elements relative to chondrite; in Figure 2B, the samples have higher levels of incompatible elements except for Y and Yb with varying Nb anomalies ($Nb^* = 0.7 \pm 0.3$). Negative Sr anomalies are common in most evolved andesite samples (e.g., NK-8). Rare plagioclase accumulation may have produced small positive Sr anomalies in some samples (e.g., CK-54). Lu/Hf values (0.07 ± 0.02) of these dikes are close to the values of ocean island basalts (0.03; Sun and McDonough, 1989).

The hypersthene-normative 65–70 Ma dolerite dikes in both central and north Kerala are tholeiitic basalts in the total alkali-silica diagram and show a trend toward Fe enrichment (Fe_2O_3 =

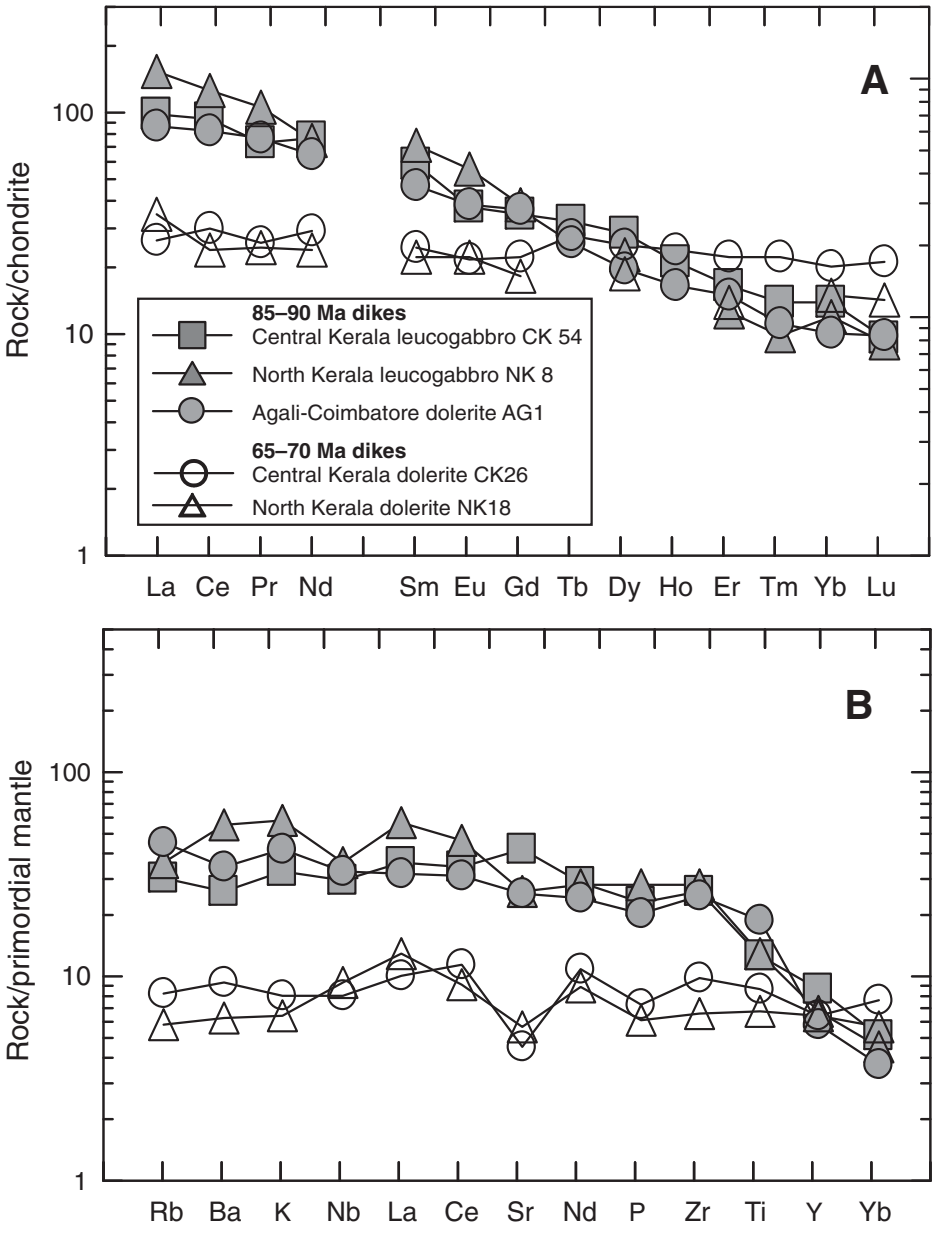


Figure 2. (A) Chondrite-normalized, rare-earth element patterns and (B) primordial-mantle–normalized, multielement diagram of the Late Cretaceous dike samples from Kerala, India. Normalization values from Sun and McDonough (1989). Open symbols are from 65 to 70 Ma dolerites and gray fillings are from 85 to 90 Ma dikes.

[1]GSA Data Repository item 2011165, Major and trace element data of 85–90 Ma (Table DR1) and 65–70 Ma (Table DR2) central/north Kerala dikes, is available at http://www.geosociety.org/pubs/ft2011.htm or by request to editing@geosociety.org.

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11–16 wt%). They have low K_2O (0.10–0.52 wt%) and P_2O_5 (0.12–0.30 wt%) similar to the values for normal Indian mid-ocean ridge basalt (MORB; http://www.petdb.org/petdbWeb/index.jsp). The dolerites also have mostly low Ni and Cr in the range of 74 to 121 ppm and 84 to 300 ppm, respectively, and moderate Y and Zr contents (12 to 37 ppm and 56 to 180 ppm; Table DR2 [see footnote 1]). Values of Zr/Y range from 2.5 to 5.7, Zr/Nb = 11–22, and La/Nb = 0.7–2.6. In Figure 2A, the dolerites have almost flat chondrite-normalized, rare-earth element patterns (~15–20 × chondritic values), while in Figure 2B, they have low incompatible element abundances relative to the 85–90 Ma samples. The 65–70 Ma dolerite dikes exhibit a familiar type of mantle-normalized pattern (Fig. 2B) that is common to many Deccan flood basalts (e.g., Mahoney, 1988) and show small positive and negative Nb anomalies (Nb* = 1.1 ± 0.3). Mild negative Sr anomaly is prominent in many samples. Their Lu/Hf values (0.17 ± 0.07) range between the MORB and ocean island basalt values but close to MORB values (0.22; Sun and McDonough, 1989).

Values of $\delta^{18}O$ for the 85–90 Ma dikes range from +3.9‰ to +8.2‰, whereas the 65–70 Ma dolerite samples have $\delta^{18}O$ values in the range +5.3‰ to +6.1‰, relative to standard mean ocean water (SMOW; Tables DR1 and DR2 [see footnote 1]). By comparison, the mean $\delta^{18}O$ value for typical MORB glasses is +5.5 ± 0.2‰ (e.g., Eiler, 2001; Cooper et al., 2004). For fresh whole-rock samples of ocean island basalt, values of $\delta^{18}O$ generally are <+5.8‰ (Kyser, 1990; Eiler, 2001). Oxygen isotope and petrographic data indicate that gabbro sample CK-15b has been modified by interaction with meteoric water, giving a low value of $\delta^{18}O$ (+3.9). Slightly low $\delta^{18}O$ value (+5.3) in the 65–70 Ma dolerite CK-26 may suggest minor modification of the oxygen isotopic composition by exchange with meteoric water. Samples affected by contamination, such as CK-50 and CK-28, have $\delta^{18}O$ >+6.6. The two 90 Ma Agali-Coimbatore samples analyzed for Sr and Nd isotopes have high Sr_i ratio (0.70479 and 0.70871), but they are less radiogenic with εNd values of +0.56 and –2.51. Their Nd model ages are 1.12 and 1.35 Ga (based on normal-MORB–type depleted mantle reservoir). The sample (44/87) has high $\delta^{18}O$ (+8.5) and low εNd (–2.5), probably indicating a crustal signature. The other sample (AG-1) has only limited crustal influence and has positive εNd (+0.56) and a $\delta^{18}O$ value of +6.6. There are some irregularities in the abundance patterns, notably in Rb, Ba, and K, suggesting that the measured concentrations may not always reflect those of the original magmas. Hence we treat these elements with caution, and wherever possible, base our petrogenetic interpretations on abundances of other elements.

Biaxial plots of both major and trace elements (Figs. 3 and 4) show that the geochemistry of the two sets of dikes, so near in age, is completely different. CaO, SiO_2, and Fe_2O_3 against MgO (Fig. 3) show that there are major differences between the 85–90 Ma and 65–70 Ma dikes. The differences are well reflected in fractionation plots using trace elements against Zr, and most of the plots in Figure 4 illustrate that the 65–70 Ma dolerite trends are consistent with some degree of crystal fractionation (e.g., MgO, Ni, TiO_2, and Y versus Zr). The set of 85–90 Ma dikes is very different and shows quite a wide scatter. Some trends (e.g., Fig. 4D, TiO_2 versus Zr) produce a striking negative correlation for the 85–90 Ma dikes, while the trend is positive for the younger 65–70 Ma dikes. In the case of correlation between Sr and Zr (Fig. 4E), the trend seems to be nearly flat and appears to be reversing beyond 400 ppm Zr. Differences between the two sets of dikes are fundamental and cannot be attributed to crystal fractionation because of the conspicuous differences in rare-earth element patterns, and also high field strength and other incompatible element ratios that are insensitive to fractionation.

Paleomagnetism

The T-χ curves (Figs. 5E and 5F) are typical of low-Ti titanomagnetite with a Curie temperature in the range of 540–560 °C. In agreement with petrographic observations, unoxidized magnetite is the dominant opaque mineral in these dikes. NRM directions are dispersed, but the ChRM directions resolved after demagnetization are well grouped for each site (Fig. 6): the 85–90 Ma dikes all cluster with northwest, moderately steep, upward directions and the 65–70 Ma dikes cluster dominantly with northwest declinations and moderate to steep upward inclinations; an antipodal ChRM is present in only three dikes.

In the ca. 85–90 Ma north Kerala dike sites (NK-14 and NK-16), thermal demagnetizations are insensitive, and a single stable component

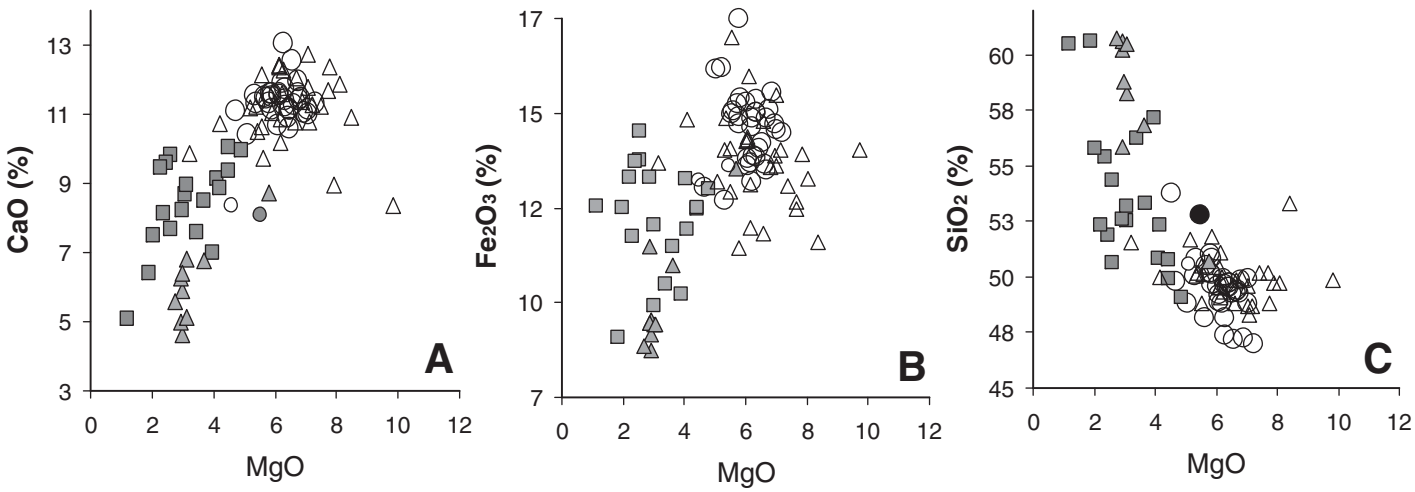


Figure 3. Diagrams showing variation in selected major elements for the dikes plotted against MgO. The 85–90 Ma dikes show lower CaO, higher SiO_2, and lower Fe_2O_3, whereas the 65–70 Ma dolerite dikes have increasing Fe_2O_3 with falling MgO. Solid symbols represent 85–90 Ma dikes (■—central Kerala leucogabbro; ▲—north Kerala leucogabbro; ●—Agali-Coimbatore dolerite), and the open symbols represent 65–70 Ma dikes (○—central Kerala dolerites; △—north Kerala dolerites).

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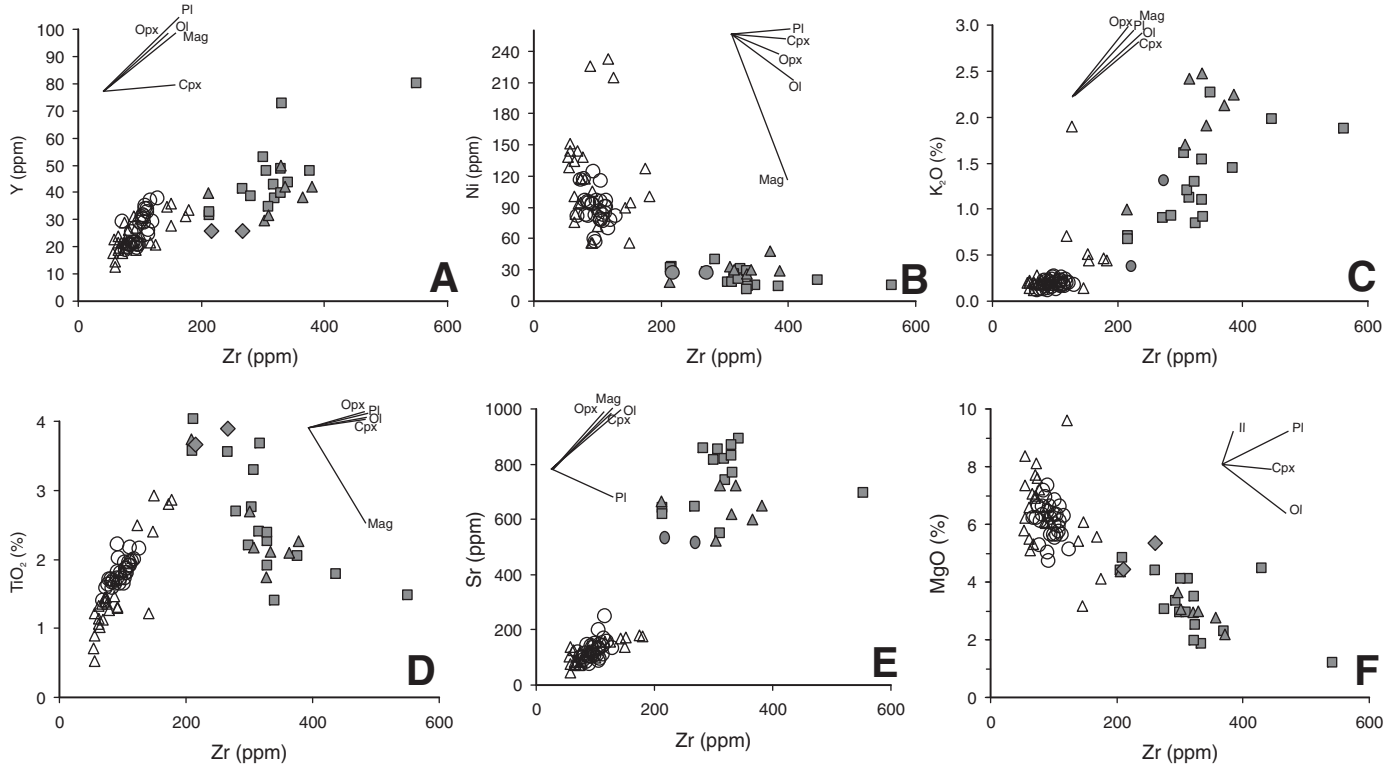


Figure 4. Elemental variations in the dikes, plotted against zirconium as an index of fractionation. Significant differences between the two groups of dikes are apparent for every element, and in addition, the fractionation trends have different directions, especially in (D). Although the dolerites are more coherent, Figure 4A also suggests two groups within the dolerites with different Y/Zr ratios. Solid symbols correspond to 85–90 Ma samples, and open symbols correspond to 65–70 Ma dike samples.

was isolated from the AF demagnetizations beyond 15 mT (e.g., Fig. 5A). Results for one sample from NK-16 and three samples from site NK-14 are widely dispersed and did not show improvement during both AF and thermal demagnetizations. Based on the data from the remaining nine samples of both sites, a single mean direction was calculated since both sites are from the same dike. At site AG-1 of the Agali-Coimbatore dolerite, four samples that yielded northwest declinations and negative inclinations shallower than reported values during AF treatment, upon thermal demagnetizations (>350 °C; Fig. 5B; AG-82), have improved the within-site grouping. However, one sample (AG-85) did not show improvement from the shallower inclinations to include into the group. The mean ChRMs estimated from the other seven samples compare well, showing coherent grouping with the other 85–90 Ma directions reported from south India (Fig. 6A). Our new results and other available pole data of 85–90 Ma igneous units from south India are unipolar, with normal polarity magnetizations, and mark Chron 34N, the Cretaceous Normal Superchron (83–117 Ma; Cande and Kent, 1995). We combined the results of ca. 85–90 Ma igneous units from all locations of south India to calculate a mean ca. 85–90 Ma paleomagnetic pole of India (Fig. 7A; Table 1). The directions reported earlier for St. Mary's Islands (Valsangker et al., 1981; also see Fig. 7A), which are dispersed and away from the recent and our present results, are probably related to retention of secondary overprints and are not included for the mean ca. 85–90 Ma pole of India. The angular standard deviation (S = 12.4) calculated from the virtual geomagnetic poles (excluding the data of Valsangker et al., 1981) is within the range (13.4 ± 0.8) predicted (McFadden and McElhinny, 1984) in the latitudes of ~30° for the period of 45–110 Ma and thus constitutes the best possible paleomagnetic pole averaging secular variations.

The ChRM directions delineated from the 65–70 Ma dolerites show a coherent within- and between-site grouping (Fig. 6B), with the α_{95} values typically <10° around the mean. The ChRM magnetizations were obtained after removal of soft (viscous) remanent magnetizations in fields between 10 and 20 mT, and the thermal demagnetizations were not useful to separate out the ChRM component (Figs. 5C and 5D). One site (NK-13) did not yield within-site coherent directions and was rejected. The ChRM directions from the dikes of north Kerala include both normal (seven dikes) and reverse (three dikes) magnetizations, in contrast to the dominant reverse polarity (five out of six dikes) from the coeval dolerite dikes in central Kerala (Radhakrishna et al., 1994). The subgroup of normal directions appears to be similar to 85–90 Ma directions owing to paleosecular variations, given the close distribution of directions of the two groups with small difference in age. The α_{95} circle of the NK-5 site is distinctive without overlapping that of the mean ChRM or that of other dikes from the area. A reversal test (McFadden and McElhinny, 1990) was performed between the bipolar characteristic remanences registered within the 65–70 Ma dolerite dikes of north Kerala, excluding the NK-5 site (7N versus 3R) and also combining bipolar remanences (7N versus 7R) of north Kerala dolerites with central Kerala dolerites of the same age. The results pass the reversal test at classification B ($\gamma = 7°–8°$).

Figure 7 compares the mean paleomagnetic pole of 65–70 Ma north Kerala dikes with the synthetic apparent polar wander path of Besse and Courtillot (2002), the superpoles of the

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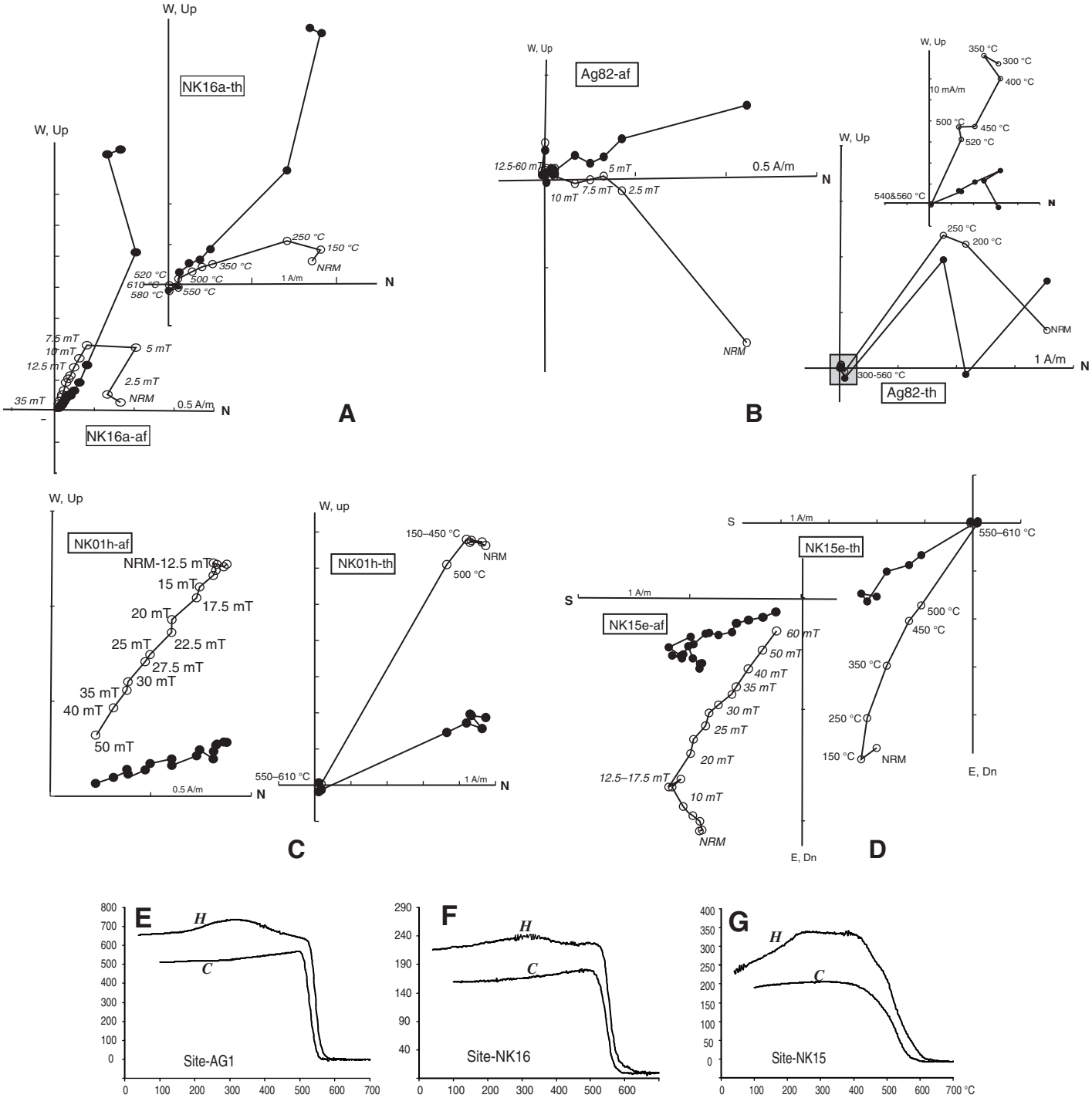


Figure 5. Orthogonal projections of magnetic remanence from representative samples of (A) 85–90 Ma dike from north Kerala-NK-16 site, (B) 85–90 Ma dike of the Agali-Coimbatore-AG-1 site and (C and D) 65–70 Ma dolerite dikes of the north Kerala-NK-01 and antipodal NK-15 sites. Suffixes "af" and "th" denote alternating field and thermal demagnetizations, respectively. Open and closed symbols denote projections onto the horizontal and vertical planes, respectively. (E–G) thermomagnetic plots depicting susceptibility-temperature variations for the sites from corresponding areas. NRM—natural remanent magnetization.

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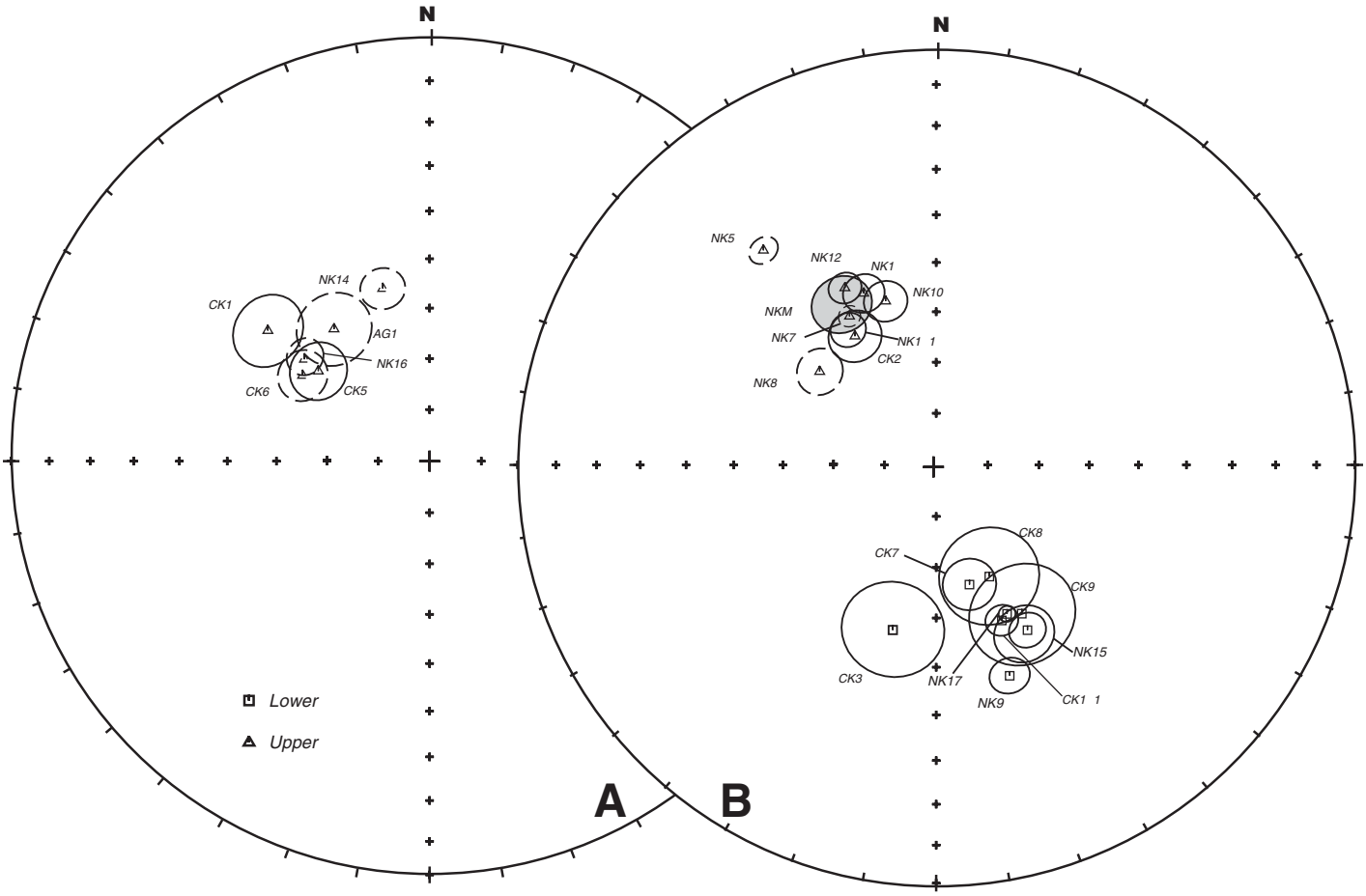


Figure 6. Equal-area projection of Late Cretaceous dikes (A: 85–90 Ma; B: 65–70 Ma) showing characteristic remanent magnetization (ChRM) of mean vectors in terms of α_{95} circles. Circles of continuous and broken lines represent lower and upper hemispheres, respectively. For comparison, our earlier results (Radhakrishna et al., 1994) from dikes of central Kerala (CK) are also plotted.

Deccan Traps along E-W Nagpur-Bombay and N-S Nasik-Mahabaleswar traverses (Vandamme et al., 1991; Chenet et al., 2009) and the central Kerala dolerites (Radhakrishna et al., 1994). The mean north Kerala pole incorporates the Deccan α_{95} circle and completely overlaps the α_{95} circles of 65 and 70 Ma synthetic poles, in conformity with isotopic age data. Similarly, the mean 85–90 Ma pole overlaps the circles of 90 Ma synthetic pole and a major part of the 80 Ma synthetic pole circle.

DISCUSSION

Fractional Crystallization and Crustal Contamination

Low Mg-value and low Ni and Cr abundances (Tables DR1 and DR2 [see footnote 1]) compared to primary melts are consistent with variable amounts of fractionation of ferromagnesian phases. The 65–70 Ma dolerites exhibit a typical tholeiitic trend of TiO_2 and Fe_2O_3 enrichment with falling MgO and increase of Zr (Fig. 4), high Ca/Al, and nearly constant SiO_2. The trends are consistent with fractionation of plagioclase, clinopyroxene, and olivine, and Fe-Ti oxides did not form a part of the fractionation assemblage. The set of dikes exhibit a decrease of Fe with Mg, variable but low Ca/Al, and variable but high SiO_2. These trends cannot be described in terms of plagioclase, clinopyroxene, and olivine fractionation within tholeiitic magma (Lightfoot et al., 1990). Typically lower Cr (<120 ppm) and Ni (<50 ppm) and high Sr (>440 ppm) values indicate greater pyroxene/plagioclase fractionation. High TiO_2 and the CaO/Al_2O_3 ratios similar to Marion Island and Funk seamount volcanic rocks suggest fractionation of clinopyroxene at higher pressures (Storey et al., 1997). However, plagioclase appears to be a dominant fractionation phase in more evolved samples (Zr >320–330 ppm). Rapid increase of SiO_2 and large variation of Ti/Zr ratios are consistent with fractional crystallization of titanomagnetite. The most evolved sample CK-16b can be generated by 62% fractional crystallization of clinopyroxene (31%), plagioclase (60%), oxide (6.5%), and apatite (1.5%) phases from CK-55, the least fractionated sample of the dikes (see Appendix). The large troughs for Ti and P in the pattern are ascribed to titanomagnetite and apatite fractionation. P_2O_5 has an overall increasing trend in the early stages with decrease of MgO, and shows inflection at 3.5 wt% MgO (or ~350 ppm Zr) where the trend reverses. Thus, more evolved samples, with higher levels of abundance patterns, have developed valleys at P.

The set of dike samples displays larger effects of crustal contamination. Some of the rocks, mostly from north Kerala, are marked by enrichment of SiO_2 and depletion of Ca/Al. Mantle-normalized patterns of the samples most influenced by crustal effects, such as CK-50 and CK-28, show stronger enrichment of Rb, K, La, Ce, and pronounced Nb troughs compared

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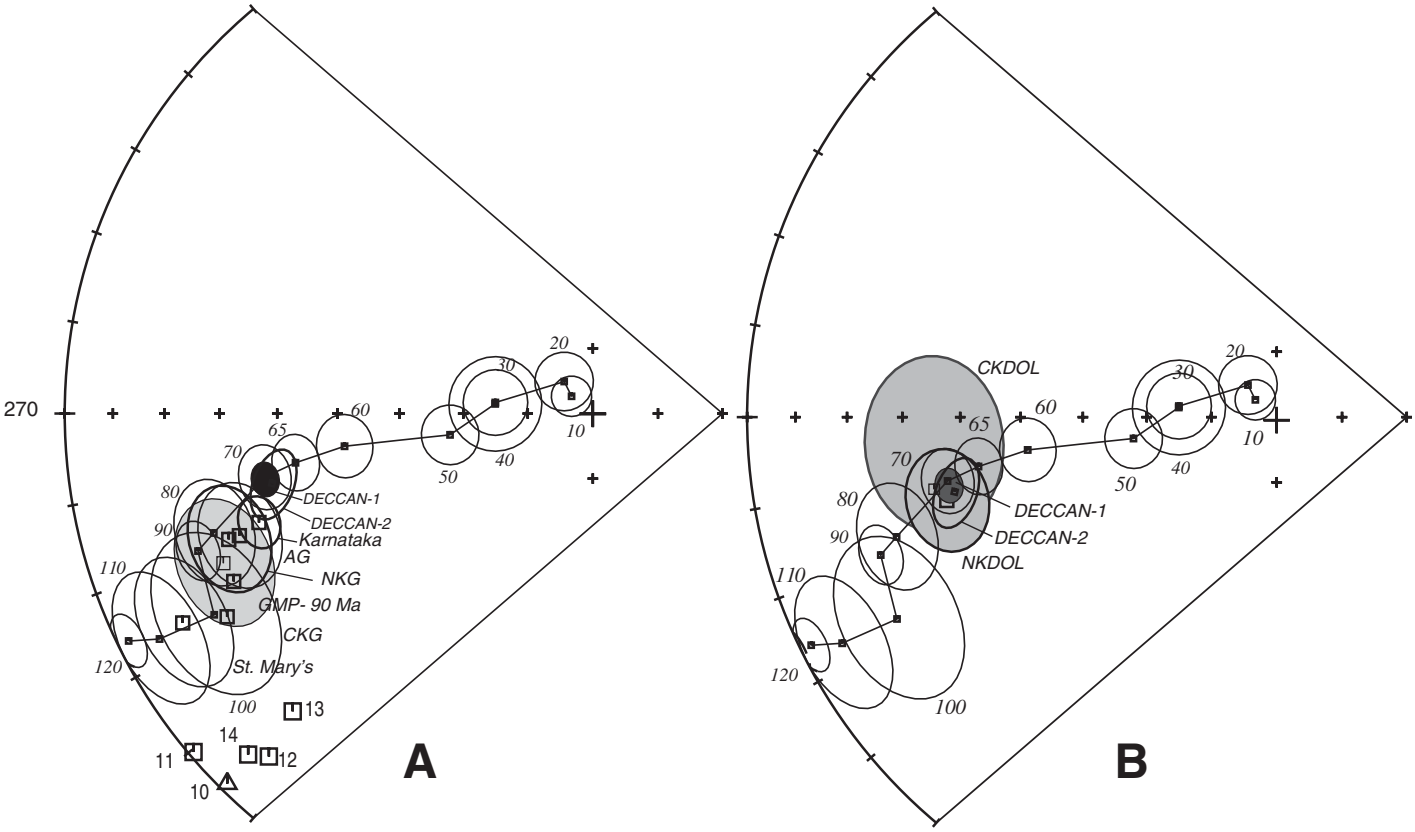


Figure 7. Mean paleomagnetic pole data of (A) 85–90 Ma dikes and (B) 65–70 Ma dikes from Kerala along with synthetic apparent polar wander path (Besse and Courtillot, 2002). In (A), the grand mean pole (GMP) of 18 sites total (85–90 Ma; Table 1) overlaps the 100–80 Ma synthetic poles. The pole data of Valsangker et al. (1981; numbered 10–14 in SW quadrant) differ from this and recent work, and suspected to retain remagnetized components are not included for calculating the mean. CKG, NKG, and AG represent the mean of central Kerala, north Kerala, and Agali-Coimbatore dike data. In (B), the mean α_{95} circle of north Kerala incorporates the Deccan superpole and completely overlaps the mean circles of 70 Ma central Kerala dolerites and 70 and 65 Ma synthetic poles. Deccan-1 is the overall mean pole of E-W Nagpur-Bombay traverse (Vandamme et al., 1991), and Deccan-2 is the overall mean pole of N-S Nasik-Mahabaleswar traverse (Chenet et al., 2009).

to the least contaminated samples. Thus high K and Rb, along with moderate Ba contents and Ce/Y and La/Th (Table DR1 [see footnote 1]), characterize the contaminant of the dikes, with the result that these ratios are slightly higher in these dikes than in a "typical" uncontaminated basaltic melt. Granulite rocks of southern India are the possible contaminants (cf. Mahoney et al., 1982). The high $\delta^{18}O$ of gabbro sample CK-28 (+8.2‰) is similar to that of granulite in southern India and Sri Lanka (~+8‰ to +8.4‰; Jiang et al., 1988; Hoernes et al., 1990) and granulite-facies xenoliths from several localities around the world (e.g., Kempton et al., 1990). The Kabbaldurga gneiss/charnockite, with high K and Rb, produces an abundance pattern fairly similar to CK-28 or CK-50 (except the troughs at Ti, which are related to titanomagnetite fractionation). Quantitative estimates (using Kabbaldurga gneiss-KD-1–119–2 of Allen and Condie, 1985) show that an addition of ~25%–30% Kabbaldurga crust to the sample CK-55, which is little contaminated, produces trace-element patterns similar to the most contaminated samples (e.g., CK-28, CK-50, or NK-19; see for details, Table DR1 [see footnote 1]). The available analyses of crustal end members are quite few and may explain some variations between the calculated and actual values.

The 65–70 Ma dolerites appear to be largely unaffected by crustal contamination, having low SiO_2, Ba/Zr (<1), Sr/Zr (<2), and La/Th (<15). Oxygen isotope data ($\delta^{18}O$ = +5.3 to +6.1; Table DR2 [see footnote 1]) also suggest limited interaction with crustal rocks. A few samples display sawtoothed, mantle-normalized patterns with pronounced Ba and Th spikes and strong depletion of Ta and Nb (CN/VA/11 and Wc; Table DR2 [see footnote 1]), cumulus mineralogy, or high K_2O (>0.50 wt%). These samples are interpreted as showing signatures of crustal contamination.

Partial Melting, Mantle Sources, and Comparison with Deccan and Madagascan Volcanism

Rare-earth elements are resistant to alteration. In the 85–90 Ma dike samples from Kerala (Fig. 2A), the pronounced enrichment in the light rare-earth elements and a steep slope between the middle and heavy rare-earth elements are consistent with small amounts of partial melting of garnet peridotite. In contrast, the 65–70 Ma dolerites have flat patterns (Fig. 2A) suggestive of more extensive melting within the spinel stability field. The Indian samples are plotted in Zr/Y versus Nb/Y diagram (Fig. 8A) adopting the melting calculations of Fitton et al. (1997; see for details of calculations, Table A1). The values of 85–90 Ma dike samples lie close to the melting curve for garnet peridotite and appear to have formed by low (~5%) mantle melting. These samples have

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TABLE 1. PALEOMAGNETIC DATA FROM LATE CRETACEOUS DIKES OF KERALA, INDIA

	Trend	Latitude (°N)	Longitude (°E)	N	N_s	D	I	k	α_{95}	λ	ϕ
65–70 Ma dikes											
North Kerala dolerites											
NK-1	NW-SE	12.04	75.40	7	7	337.51	–54.07	240	3.9	38.7	279.2
NK-5	NW-SE	12.39	75.27	6	6	321.23	–35.38	520	2.6	40.2	305.9
NK-7	NW-SE	11.87	75.55	3	3	329.98	–56.95	3479	21	33.1	263.9
NK-8	NW-SE	11.91	75.78	7	7	308.40	–61.57	184	4.5	17.9	293.0
NK-9	NW-SE	11.74	75.47	6	6	159.94	45.45	330	3.7	46.7	102.0
NK-10	NW-SE	11.88	75.54	7	7	343.35	–57.08	218	4.1	38.0	272.3
NK-11	NW-SE	11.87	75.52	6	6	327.13	–59.18	380	3.4	29.7	284.2
NK-12	NW-SE	11.94	75.48	8	8	333.04	–51.53	323	3.1	38.9	285.0
NK-13	NW-SE	11.72	75.48	5	5	----------No grouping----------					
NK-15	NW-SE	11.15	75.18	6	6	150.14	52.54	304	3.5	37.4	107.5
NK-17	NW-SE	11.44	75.79	4	4	153.42	57.35	3204	1.6	34.9	101.0
Mean of 10 sites (dikes)						330.6	–53.5	72	5.7	36	283
Grand mean pole of Deccan on E-W Nagpur-Bombay traverse (Vandamme et al., 1991); λ = 36.9N; ϕ = 281.3E; A_{95} = 2.4											
Grand mean pole of Deccan on N-S Nasik-Mahabaleswar traverse (Chenet et al., 2009); λ = 37.8N; ϕ = 282.6E; dp/dm = 3.2–5.1											
85–90 Ma dikes											
North Kerala											
NK-14	NW-SE	11.36	75.82	6	6 (3)	346.38	–54.79	865	4.2	41.6	270.7
NK-16	NW-SE	11.41	75.78	6	6	308.00	–58.90	258	4.2	19.8	295.9
					9	321.66	–58.85	52	7.2	27.8	288.5
Agali-Coimbatore sector											
AG-1	ENE-WSW	11.11	76.74	8	7	324.06	–57.98	72	7.2	30	289
Central Kerala (Radhakrishna et al., 1994)											
CK-1	NNW-SSE	9.93	76.6	8	8	308	–48	66	6.9	26.8	307.2
CK-5	NNW-SSE	9.76	76.66	7	7	309	–62	117	5.6	19.9	293.5
CK-6	NNW-SSE	9.88	76.61	6	6	304	–60	241	4.9	18.0	297.7
Mean of five sites of two Dharwar dikes (Kumar et al., 1988)						329	–54	509	3.4	34	288
Mean of seven sites of St. Mary's Islands volcanics (Torsvik et al., 2001)						301.2	–58.5		7.5	14	297
Grand mean pole calculated from all 18 sites available from south India						λ = 24; ϕ = 293; A_{95} = 5.9					

Note: N—number of samples collected; N_s—number of samples obtained by characteristic remanent magnetization (ChrRM); D/I—declination and inclination (in degrees); α_{95}—95% confidence circle; k—precision parameter; λ and ϕ are pole latitude and longitude, respectively. CK-1, -5, and -6 are from Radhakrishna et al. (1994).

high chondrite-normalized Tb/Yb_n ratios (>2), indicating the presence of garnet in the source region. The low Sc values (<25–30 ppm) also may reflect deep, garnet-present melting. Clinopyroxene fractionation influences the Sc contents, but the low values may not be attributed to clinopyroxene fractionation because the bulk distribution coefficients of fractionation phases are likely to be about unity. The 65–70 Ma dolerite samples, with Sc concentrations in excess of ~30 ppm corresponding to shallow, spinel peridotite melting, generally plot below the garnet and spinel peridotite melting curves, at relatively low values of Nb/Y and Zr/Y suggesting 10%–20% melting in the garnet-spinel transition zone (Fig. 8A).

Rare-earth elements have been further used for testifying the inferences of melting and for correlations with Madagascan volcanism. Intercepts on melting curves in a Ce/Sm_n versus Yb/Sm_n diagram (Fig. 8C) suggest that the 85–90 Ma dikes of Kerala, like the Mg-Ti tholeiites of Mananjary or a few Tamatave volcanic rocks on the east coast of Madagascar, are largely the melts produced by ~5% melting of garnet lherzolite. Similarly, these dikes in a La/Sm_n versus Tb/Yb_n diagram (Fig. 8D) also may constitute ~5% melts of garnet lherzolite with minor addition of a crust-derived component (~2%) to a light rare-earth–depleted source. The tholeiitic (THO) and transitional (TR) series of volcanic rocks from the Mailaka and Bemaraha areas of central-western Madagascar are distributed spatially in the region of low-degree melts of spinel lherzolite. This petrogenetic inference is in agreement with the regional observation that many volcanic types in the eastern part of Madagascar, in contrast to the central and western Madagascan units, require significant melting in the garnet stability field (Melluso et al., 2001) and the volcanics in these regions are compositionally distinctive (Mahoney et al., 2008).

The 85–90 Ma samples have high Zr/Y and Nb/Y relative to lavas from the upper formations of the Deccan Traps and are similar in composition to many Cretaceous tholeiites from eastern Madagascar (Fig. 8B; Dostal et al., 1992; Storey et al., 1997). Their Zr/Nb (11–25) and Hf/Ta (3.1–9.1) are comparable with the values observed in basalts along the east coast of Madagascar (Storey et al., 1997). The similarity of these rocks with the east coast tholeiites of Madagascar is more clearly pronounced in incompatible element patterns (Fig. 9A). Similarly, recent study by Melluso et al. (2009) has shown that the St. Mary's Islands volcanic rocks on the southwest coast of India are geochemical analogs of the eastern Madagascar volcanics. Coeval dikes from Karnataka (Kumar et al., 2001) were not analyzed for Nb and Ta, but the patterns of other elements differ considerably from the Kerala dikes. Their patterns appear to be close to those of rocks of Sambava area and some rocks of the Tamatave area on the east coast of Madagascar (Storey et al., 1997). Prior to seafloor spreading in the Mascarene Basin, west of India, southern India was juxtaposed with the eastern margin of Madagascar (Storey et al., 1995). The 85–90 Ma Kerala dikes are similar in age to the analyzed Cretaceous Madagascan lavas, the bulk of which appear to have been erupted above the Marion (Prince Edward) plume at ~87–90 Ma (Storey et al., 1995, 1997).

In Figure 8B, some of the 65–70 Ma samples plot along or below the lower bound of the Iceland array (i.e., they are relatively low in Nb), and have Zr/Y ratios comparable to those of T-MORB (transition mid-ocean ridge basalt) of the Southwest Indian Ridge (illustrated in fig. 2 of Fitton, 2007). These samples could have been produced by melting of a mixed source, consisting of both plume and Indian MORB mantle components. In this regard, they are similar to certain enriched-MORB–like, upper-

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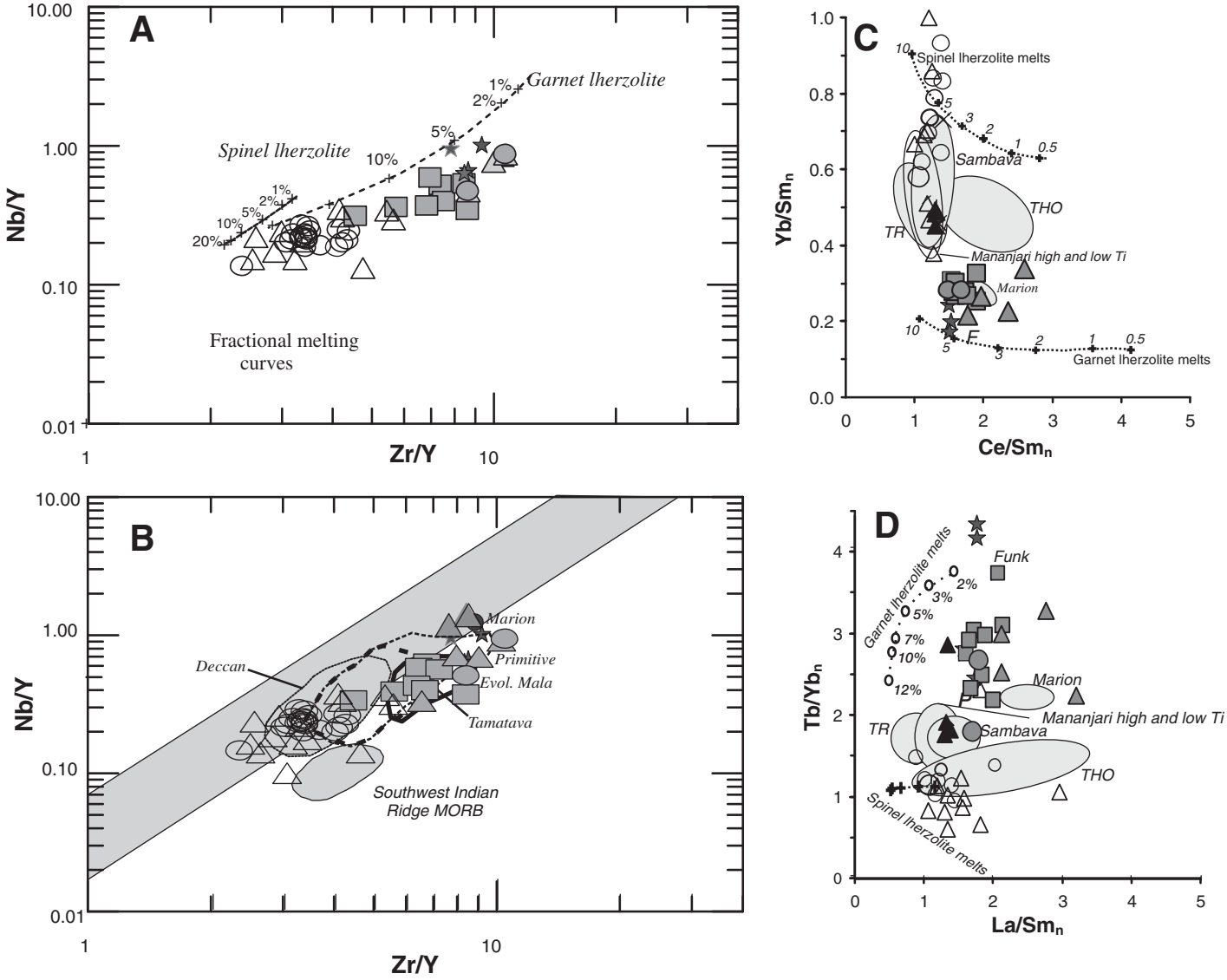


Figure 8. (A) Nb/Y versus Zr/Y plot for samples of 85–90 Ma and 65–70 Ma dikes compared with Madagascan ferrotholeiites (data of Storey et al., 1997; Melluso et al., 2001). Fractional melting curves are shown for garnet and spinel lherzolite, calculated according to the procedures of Fitton et al. (1997). Melt parameters and partition coefficients are given in Appendix. (B) Diagram comparing the dikes and Madagascan basalts with the upper and lower bounds on data obtained for lavas from the Iceland Neovolcanic zone (Fitton et al., 1997). Fields for Southwest Indian Ridge mid-ocean ridge basalt (MORB) (Fitton, 2007) and Wai subgroup basalts from the Deccan Traps (data sources in Mahoney, 1988) are shown for reference. Note that data for the Cretaceous dikes from the Seychelles (Devey and Stephens, 1991, 1992) generally plot within the field of Wai subgroup lavas. (C) The variation of Ce/Sm$_n$ versus Yb/Sm$_n$ for Kerala dikes in comparison with Madagascan volcanics. The compositions of aggregated melts produced by different degrees of melting of a spinel lherzolite and garnet lherzolite sources are shown. Details of model calculations are as given in Storey et al. (1997). Solid stars and triangles are high Mg-Ti and Tamatave basalts of Madagascar; Prince Edward Island (P) and Funk seamount (F) are from Storey et al. (1997). (D) The variation of La/Sm$_n$ versus Tb/Yb$_n$ diagram for Kerala dikes in comparison with Madagascan volcanic rocks. Details of melting curves and calculations are as given in Melluso et al. (2001). Data symbols are the same as in Figure 2.

level Deccan lavas (particularly those of the Wai subgroup, Fig. 9B). The T-MORB characteristics of the dolerites are comparable to those of 65–70 Ma Ambenali and Panhala flows of the Deccan near Mahabaleswar (e.g., Zr/Nb >11; Sr <240 ppm; Ba <100 ppm; Rb <4 ppm; see Lightfoot et al., 1990, and also Widdowson et al., 2000) and differ significantly from the Toranmal and Pavagadh basalts (Zr/Nb <10; Sr and Ba values of 400–500 ppm; Rb >15 ppm; and also high Nb values of >20 ppm; Mahoney et al., 2000; Sheth and Melluso, 2008). Compared to the upper Deccan flows (Ambenali and Panhala), a minor negative Nb anomaly is seen in some Kerala dolerites. Such anomalies are explained in terms of crustal contamination or lithosphere mantle sources preserving a subduction component. Isotopic data used to identify mantle lithosphere sources for some of the southwestern Deccan flows (Lightfoot and Hawkesworth, 1988), and more recently in other low-Ti picritic

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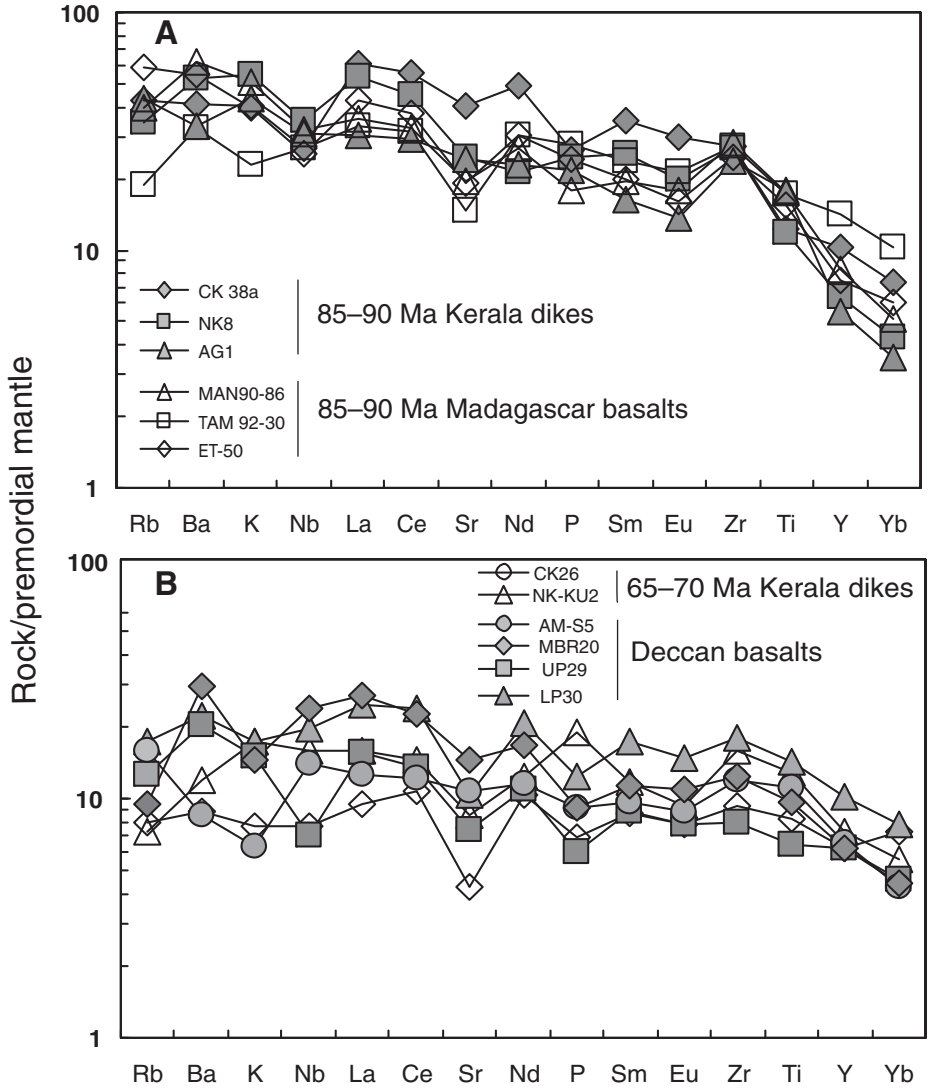


Figure 9. (A) Primordial-mantle–normalized, multielement plot of 85–90 Ma dikes in comparison to east coast basalts of Madagascar and (B) 65–70 Ma Kerala dikes in comparison to upper stratigraphic (Wai Group) Deccan and high titanium Madagascar tholeiite. Eastern Madagascar samples Mananjary (MAN) and Tamatave (TAM) are from Storey et al. (1997), and the evolved tholeiite (ET) is from Dostal et al. (1992). The Deccan samples of lower Poladpur (LP), upper Poladpur (UP), and Mahabaleswar (MBR) formation are from Lightfoot and Hawkesworth (1988), and the sample of Ambenali (AM) formation is from Widdowson et al. (2000).

basalts in Gujarat (Melluso et al., 2006), are not presently available for central and north Kerala dolerites. However, some dike samples are displaced to low Zr/Nb values with little change in Rb/Nb and Th/Ta values similar to some of the upper level Deccan lavas. Such trends are difficult to model by crustal contamination and are better attributed to lithosphere mantle end member (Lightfoot and Hawkesworth, 1988; Lightfoot et al., 1990). Decompression and strong attenuation along the continental margin of India as a result of rising Reunion plume may have supplied thermal energy for the lithosphere contribution to magmatism in southern India, either immediately preceding or accompanying eruption of the bulk of the Deccan lavas.

Implications for Indian Ocean Mantle Plumes and Tectonic Reconstructions

In the absence of precise isotopic age data and geochemical data from Madagascar, the coexistence of the 85–90 Ma and the 65–70 Ma Indian dikes, together with 65–68 Ma ages for Deccan flood basalts, were interpreted to relate to incubation of the Reunion plume for over 10 Ma (Radhakrishna et al., 1994). However, the recent precise geochronology and geochemical results allow direct comparisons between the Late Cretaceous igneous activity in Madagascar and south India. Such comparisons, as illustrated in the previous section, suggest that the 85–90 Ma dike magmatism in India and Madagascan lavas belong to the same igneous episode related to the Marion (47°S: 37.8°E) plume. Ernst and Buchan (1997) suggested that the 85–90 Ma Indian dikes belong to the Marion plume event. In view of this, the paleomagnetically derived paleolatitudes have important implications for the ongoing stationary versus migrating mantle plume debate.

The paleolatitude estimates for the southern tip of India (8.1°N: 78.3°E) determined from the paleomagnetic results of 90 Ma Cretaceous dikes from three different areas in Kerala (32–46°S; mean: 40.9°S) are nearly identical and very much comparable with the paleolatitude estimates (35–43°S; mean: 39.4°S) from dikes of the same age from a different area in the Karnataka craton (Table 1). The data are also within the broad range (29–57°S) reported for the 90 Ma St. Mary's Islands volcanic rocks (Torsvik et al., 2000). A paleolatitude of 42.8 +5.3/–10.0°S is calculated for the southern tip of India from the combined data of 18 sites. However, two sites from a study by Torsvik et al. (2000) have yielded exceedingly high paleolatitudes (beyond 70°S) compared to the combined paleomagnetic sites from both Madagascar and India. Excluding these two sites, the remaining 16 sites yield a paleolatitude of 41.2 +5.2/–4.5°S for the southern tip of India.

There are two possible explanations for these paleolatitudes in relation to Marion plume activity. (1) It is likely that the Marion plume is situated at its present location (47°S: 37.8°E), under southwest Madagascar in the 90 Ma reconstruction, as suggested from Madagascan data (Torsvik et al., 1998; Riisager et al., 2001). In such a case, the Kerala region of southern India might have been situated on the outer edge of the plume. Considering that Marion-related volcanism has a spatial spread in the whole of Madagascar, particularly in the northern extreme, spanning for ~12° in radial distance, and the spread of the volcanism up to the St. Mary's Islands off India, the plume head should be unacceptably larger than the estimated 1000–2000-km-diameter (White and McKenzie, 1989), if the plume was situated at its present latitudinal location at ca. 90 Ma. (2) The published true polar wander (TPW) curves (Besse and Courtillot, 2002) produce little offset at 85–90 Ma period (fig. 4, O'Neill et al., 2003; fig. 2, Courtillot et al., 2003). Hence,

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alternatively, it is more reasonable that the plume was situated ~5° north to its present location during the ca. 90 Ma volcanism as indicated by the mean paleolatitude estimates of this study. The spread of paleolatitudes of both Indian and Madagascan Late Cretaceous igneous units is well within the predicted (1000–2000 km) diameter of a plume head, and this scenario suggests southward migration of Marion plume. These interpretations are compatible with the prediction of global mantle-circulation models of ~7° southward migration for the Marion plume (O'Neill et al., 2003; Steinberger and O'Connel, 1998; Steinberger et al., 2004). The paleolatitudes from southern Madagascar may be explained by either eruption of magmas over the southern edge of the plume or melt propagation and distal eruption from the central plume source. Or, the Marion plume may have been northward beneath central Madagascar at ca. 90 Ma and then beneath southwest Madagascar at 83 Ma as suggested by O'Neill et al. (2003).

Our present paleomagnetic results on 65–70 Ma dikes of Kerala are in good agreement with results on the Deccan Traps (Vandamme et al., 1991; Chenet et al., 2009); the paleolatitude estimates of 65–70 Ma Kerala dikes (27.4 +5.7/–5.0°S; Nagpur in central India) and the Deccan Traps (28.4 ± 2.4°S on E-W Nagpur-Bombay traverse and 27 ± 3.2°S on N-S Nasik-Mahabaleswar traverse) are consistent with the suggestion of a northward migration of Reunion plume predicted from the global mantle-circulation models. However, published TPW curves may also account for 5°–6° northward motion, and it is difficult to distinguish between the hotspot motion and the TPW contribution. The superpole of Rajmahal Traps (Basu Mallik et al., 1999) supports an already advocated southward migration of the Kerguelen plume. The latitudinal migration for these Indian Ocean plumes, as suggested from paleomagnetic data, is in agreement with the estimates based on global mantle-circulation models (Steinberger et al., 1998, 2004; O'Neill et al., 2003).

Paleomagnetic results, particularly for the 85–90 Ma igneous units are also useful to test the Madagascar-India linkage in Gondwana reconstructions. Different paleogeographic reconstructions are available in the literature (see Melluso et al., 2009, for details). Our present paleolatitude estimates for the southern tip of India (41°S), and the paleolatitude estimates reported (Torsvik et al., 1998; Riisager et al., 2001) for the southern tip of Madagascar (47°S), fit southern India against northern Madagascar as shown in Figure 10. Such a reconstruction is consistent with the fit positions recently suggested based on plate kinematic models and satellite-derived gravity data (Eagles and Konig, 2008). In the proposed reconstruction, the Ranotsara-Bongolava shear zone in southern Madagascar, as proposed by some workers, cannot be extended across into the Achankovil "shear zone" in the southwestern part of India, or

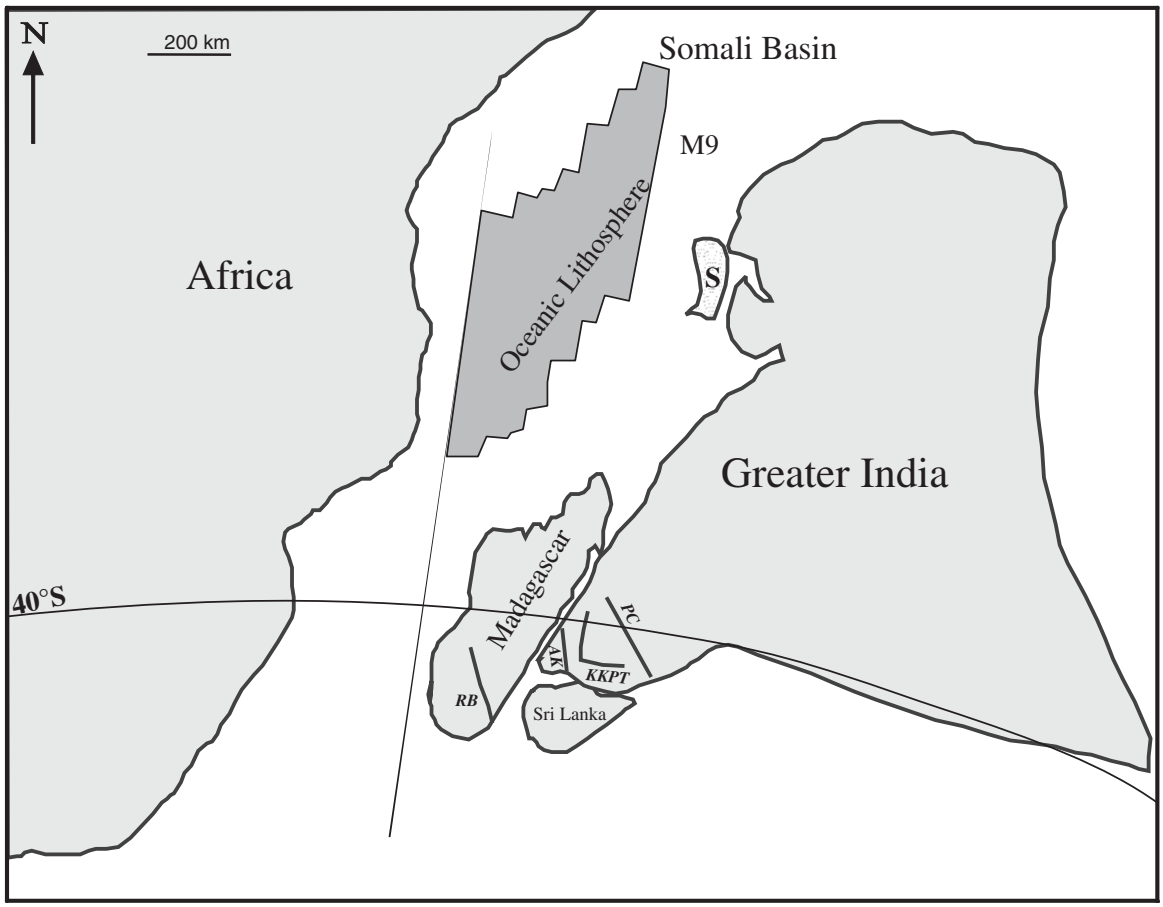


Figure 10. Late Cretaceous (ca. 90 Ma) reconstruction (modified after Torsvik et al., 1998). Madagascar and India as indicated by the paleomagnetic data of present work. The designated Ranotsara Bongolava (RB), Achankovil (AK), Palghat-Cauvery (PC), and Karur-Kambam-Painavu-Trichur (KKPT) shear zones are shown. S denotes Seychelles. Details in text.

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into the Palghat Cauvery shear zone to the north, representing a major crustal-level geotectonic boundary for Gondwana amalgamation, as suggested by us earlier (Radhakrishna et al., 1990; Radhakrishna et al., 1999; see also Radhakrishna and Joseph, 1996). While reinforcing the lack of distinct crustal history across Palghat-Cauvery and that the Achankovil may constitute an attenuated limb of a major isoclinal fold, a recent structural and U-Pb-Th zircon and monazite geochronology study (Ghosh et al., 2004) suggested that the Karur-Kumbam-Painavu-Trichur shear zone farther north of Achankovil shear zone was a terrane boundary and that it constitutes the Ranotsara-Bongolava extension in south India. However, the reconstruction in Figure 10 does not support the extension of Ranotsara-Bongolava zone of Madagascar into south India as a major terrane boundary, and such a terrane boundary can be traced only to the south of the Indian shield, probably in the microcontinental slivers fragmented from the Indian shield during the Late Cretaceous.

CONCLUSIONS

Late Cretaceous igneous activity at 85–90 Ma and 65–70 Ma is manifested as dike magmatism in central Kerala, north Kerala, and the Agali-Coimbatore regions on the west coast of India. New geochemical and paleomagnetic results coupled with previously published isotopic age data clearly distinguish both sets of dikes. The set of dikes may constitute part of the Madagascan continental flood basalt with similar ages and major- and trace-element geochemistry. Paleomagnetic data provide an improved and better constrained 90 Ma pole for greater India. The 65–70 Ma dolerite dikes closely compare in age and geochemistry with the upper formations of the Deccan flood basalts and tholeiitic dikes in Seychelles. Their paleomagnetic pole compares well with the Deccan superpole. We relate the distinct dike sets to the Marion plume, and Reunion plume and breakup of India-Madagascar and the India-Seychelles microcontinent. While the geochemical data traced the origin within garnet peridotite and spinel peridotite mantle sources for the 85–90 Ma and 65–70 Ma set of dikes, respectively, the paleomagnetic data place bounds on Marion and Reunion hotspot motions and paleopositions of India-Madagascar in predrift configuration.

The paleolatitude estimates of the 85–90 Ma dikes suggest, in conformity with the predictions of global mantle-circulation models (O'Neill et al., 2003), a southward migration of the Marion plume. At the same time, the paleolatitude of the 65–70 Ma dolerites, like the paleolatitudes of the Deccan Traps, can be interpreted as suggesting a northward motion of the Reunion plume as predicted by global mantle-circulation models. However, taking into account published TPW curves (Besse and Courtillot, 2002), it is difficult to distinguish between the TPW contribution and hotspot motion. Finally, the paleolatitudes place southern India adjacent to the northern Madagascar in the predrift continental assembly and do not support many reconstructions that correlate the "shear" zones across India and Madagascar.

APPENDIX

Fractional Crystallization

In the 90–85 Ma dikes, samples CK55 and CK16b, with little crustal effects, represent the most extreme compositions by fractionation crystallization. The most fractionated sample CK16b is modeled to have formed from least fractionated sample CK55 by 62% fractional crystallization of clinopyroxene (31%), plagioclase (60%), oxides (6.5%), and apatite (1.5%). The modeled and measured values are shown as mantle-normalized patterns below (Fig. A1). Partition coefficients are taken from the compilations of Rollinson (1993) and Ewart et al. (2004) for model calculations. Note a good match in the shape of the patterns, except for P and Ti. Troughs for these elements in measured values of CK16b can be explained by fractionation of small amounts of titanomagnetite and apatite. Major-element mixing calculations are also compatible with titanomagnetite (~6%) and apatite (~1.5%) extraction.

Contamination

The saw-toothed patterns (Fig. A2) of the most-contaminated samples (e.g., CK28) appear to have developed by bulk contamination of magmas with South Indian crustal units. The overall shape of the pattern is reproduced by addition of 25%–30% Kabbaldurga gneiss/charnockite–type crust to a least-contaminated dike sample (CK55). The difference in measured values of sample CK28 is marked by a trough at Ti, which is attributed to fractionation of titanomagnetite, and which is evident in these samples. The available analyses of crustal end members are quite few and may have resulted in some differences in the abundance levels, although the shapes of measured and calculated patterns are similar.

TABLE A1. THE PARTITION COEFFICIENTS AND MANTLE AND MELT PARAMETERS USED FOR MELT CALCULATIONS

Mineral/melt	Partition coefficients			Mantle mode		Melt mode	
	Zr	Nb	Y	Sp-lhz	Gt-lhz	Melt (sp)	Melt (gt)
Olivine	0.0013	0.000041	0.007	0.576	0.5974	0.1	0.03
Orthopyroxene	0.013	0.0001	0.025	0.266	0.2104	0.2	0.03
Clinopyroxene	0.128	0.007	0.421	0.1054	0.0672	0.68	0.44
Spinel	0.4	0.4	0.0078	0.0326	0	0.02	0
Garnet	0.27	0.0042	2.8	0	0.105	0	0.5
Melt	1.0	1.0	1.0	0.02	0.02	0	0

Note: Mantle and melt parameters are same as those used by Fitton et al. (1997) and the partition coefficients are taken from Donnelly et al. (2004), Eiler et al. (2005), and Melluso et al. (2005). Sp—spinel; gt—garnet; lhz—lherzolite.

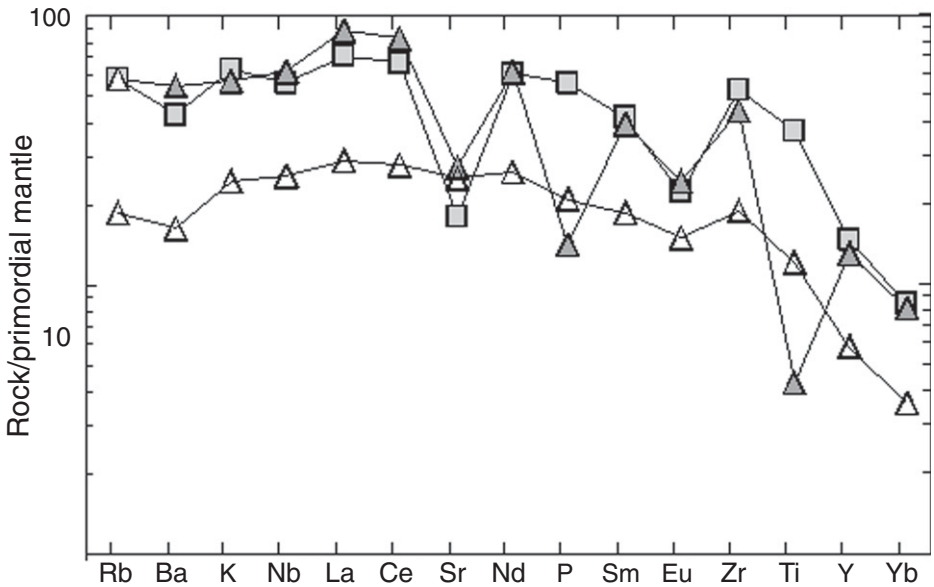


Figure A1. Primordial mantle–normalized incompatible element patterns of observed and calculated values of 90–85 Ma dike samples from central Kerala, India. Symbols: △—least fractionated sample CK55; ▲—observed values of most fractionated sample CK16b; ■—calculated values by fractionation model for CK16b.

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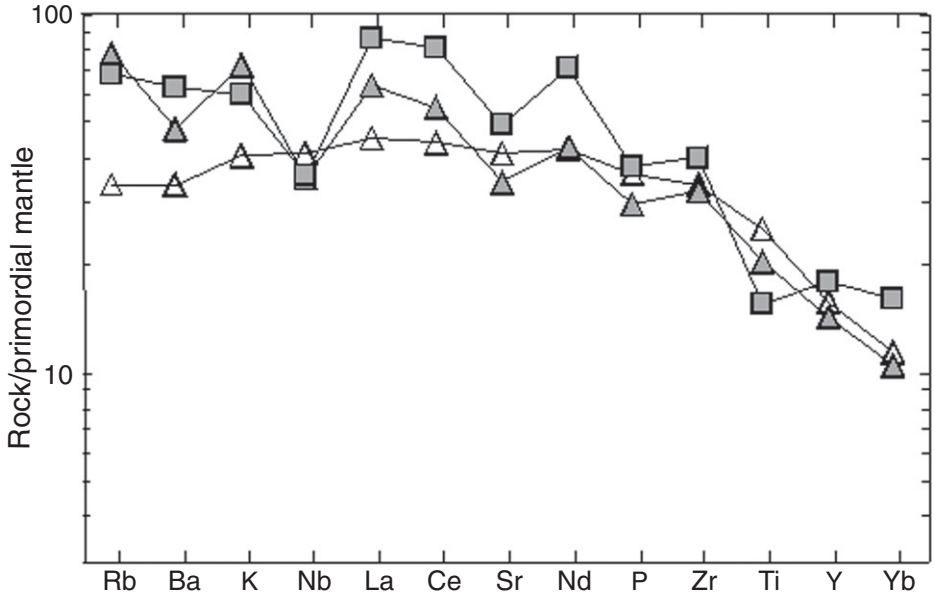


Figure A2. Mantle-normalized incompatible element patterns of little-contaminated sample CK55 (△), along with observed values (▲) of most-contaminated sample CK28 compared to calculated values (■) by addition of 25% crust (Kabbaldurga gneiss: KD-1–119–2) to sample CK55.

Melt Parameters and Partition Coefficients Used for Zr/Y versus Nb/Y Melt Calculations in Figure 8

The partition coefficients and mantle and melt parameters used for melt calculations are given in Table A1. Mantle and melt parameters are the same as those used by Fitton et al. (1997). The partition coefficients are taken from Donnelly et al. (2004), Eiler et al., (2005), and Melluso et al. (2005).

ACKNOWLEDGMENTS

T.R. was supported by grants from the Indian National Science Academy and the Department of Science and Technology (DST), New Delhi (grants ES/23/009/86 and ESS/CA/A9-11/92). The critical comments by Drs. J.J. Mahoney and L. Melluso on an earlier version of the manuscript have considerably helped to improve the manuscript. The Indo-French Center for Promotion of Advanced Research (IFCPAR) grants are the motivation for our renewed interest in the Cretaceous dikes. The isotope work was carried out during the tenure of a Better Opportunities for Young Scientists in the Chosen Areas of Science and Technology (BOYSCAST) fellowship funded by the DST. We thank D.B. Wenner for providing isotope facilities at the University of Georgia, and V. Balaram and G. Balasubramonian for analytical work. We are grateful to the Director, Center for Earth Science Studies, for permission to carry out this research.

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Science Editor: Brendan Murphy
Associate Editor: Richard Ernst

Manuscript Received 5 March 2010
Revised Manuscript Received 10 October 2010
Manuscript Accepted 14 October 2010

Printed in the USA

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